

EOPC2202 ANALOG & DIGITAL ELECTRONICS CIRCUIT LABORATORY (0-0-3)

Course Objective:

1. To understand the operation of the various bias circuits of BJT, JFET and MOSFET.
2. To study the frequency response of BJT, JFET and MOSFET-amplifier.
3. To analyse RC phase shift oscillator and square wave testing of amplifier.
4. To analyse combinational Logic circuits using Karnaugh Maps or Quine McClusky Techniques.
5. To study Decoders, Encoders, Digital multiplexers, Binary comparators, and Flip-Flops.
6. To design and analyse Synchronous and Asynchronous Sequential circuits, registers and Counters.

List of Experiments (At least 10 experiments)

Analog Electronics circuit Experiments (Any Five)

1. Design and simulate voltage divider biasing circuits using BJT-CE configuration and compare the results.
2. Design and simulate of voltage divider biasing circuits using JFEET-CS configuration.
3. Design and simulate of Self-biasing circuits of MOSFET.
4. Determine the frequency response of BJT CE- amplifier: low frequency, Mid-frequency and high frequency response.
5. Determine the frequency response of BJT emitter follower (CC-amplifier) circuit.
6. Determine the frequency response of JFET CS- amplifier Circuit.
7. Determine the frequency response of MOSFET CS- amplifier circuit.
8. Study of RC phase shift oscillator using BJT/OPAMP.
9. Study of Class A, B power amplifier.

Digital Electronics circuit Experiments (Any Five)

1. Design of Binary Multiplier.
2. Design and implementation of code converters gray to binary and BCD to seven segmentdisplay.
3. Design and implementation of a function using Multiplexer (MUX) and De-multiplexer(DEMUX).
4. Design of function using decoder and encoders.
5. Flip-Flop: assemble, test and investigate operations of S-R, D and J-K flip-flops.
6. Shift Registers: Design and investigate the operation of all types of shift registers with Parallel load.
7. Counters: design, assemble and test various ripple and synchronous counters - decimal counter, binary counter with parallel load.

Course Outcome: After completion of the sessional student should be able to

1. Understand about the operation of the various bias circuits of BJT, JFET and MOSFET.
2. Acquire knowledge on the frequency response of different amplifier circuits.
3. Learn the operation of combinational Logic circuits using Karnaugh Maps or Quine McClusky Techniques.
4. Study Decoders, Encoders, Digital multiplexers, Binary comparators, and Flip-Flops.
5. Design and analyse Synchronous and Asynchronous Sequential circuits, registers and Counters.